

Surfaces with  $k=0$

§1 Compact complex surf / compact Kähler / projective.

See [BHPV] §IV 3.6.

Thm ([BHPV] Thm 3.1 or [Bea] App B)

compact complex surf is Kähler (i.e.  $\exists$  closed Hermitian form)  
iff  $b_1 = \text{even}$ .

pf Analytic

Cor

K3, Complex tori are Kähler.

⊗ Complex tori : standard metric on  $\mathbb{C}^2 (\leadsto \mathbb{C}^2/\Lambda)$

⊗ K3 : See also Siu (1983) "Every K3-surface is Kähler"

Thm ([BHPV] Thm 6.2 or [Bea] Prop B.2)

compact surf is proj iff  $\exists L : \text{line bundle w/ } c_1(L)^2 > 0$ .  
L: big

Cor

gen type is proj

$K_X$ : big

⊗ For curve

proj = Kähler

$\Rightarrow$  ruled = proj

$P(E)$

$\downarrow$  proj  
 $\subset$  proj

Summary

cpx cpt surf

cpt Kähler

Hopf (ellip)  
Inoue  
Kodaira (ellip)

non-alg K3 (ellip)  
cpx tori

proj

rational / ruled  
pol. K3 / Ab / Enr. / biellip  
Some Ellip surf.  
gen type.

We will now proceed in a setting where Hodge theory applies!

## §2 Classical invariants

[2]

Rough idea

①  $\text{Pic}(X) = \{\text{line bundle}\} / \cong$

$\downarrow$   
 $L \mapsto L^* := L \setminus \{\text{zero section}\}$  is  $\overset{\circ}{G}_m$ -torsor  
 $(\mathcal{O}_X^*)$

if  $L^*$  has a global section ( $\Rightarrow$  nowhere vanishing section)  
 $\Rightarrow L \cong \mathcal{O}_X$

So we get  $\text{Pic}(X) = H^1(X, G_m) (= H^1(X, \mathcal{O}_X^*))$

② Consider

$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 1$  exact  
 $x \mapsto e^{2\pi i x}$

$\Rightarrow H^1(X, \mathcal{O}_X) \rightarrow H^1(X, \mathcal{O}_X^*) \rightarrow H^2(X, \mathbb{Z})$

$\downarrow \quad \quad \quad \downarrow$   
 $L \xrightarrow{\text{conn hom}} [\text{div}(s)] \quad 0 \neq s \in H^0(X, L)$

(another choice  $\rightarrow \text{div}(s) = \text{div}(s) + (f)$ )

③ Lefschetz (1.1) - Thm

$\text{im}(C_1) = H^2(X, \mathbb{Z}) \cap H^{1,1}(X) =: NS(X)$

pf see Griffiths, Harris or Wiki.

④  $\ker(C_1) = \text{Pic}^0(X)$  : conn comp of the identity

Fact  $\cdot \text{Pic}^0(X)$  : Ab. var (so smooth)

$\cdot T_0 \text{Pic}^0(X) = \{\text{infinitesimal def}\}$   
 $= H^1(X, \text{End}(L)) = H^1(X, \mathcal{O}_X)$

$\cdot \text{Pic}^0(X) = (\text{Alb}(X))^v$

$\cdot NS(X) = \text{Pic}(X) / \text{Pic}^0(X) = \text{Div}(X) / \sim_{\text{alg}}$

$\dim \text{Pic}^0(X) = h^1(X, \mathcal{O}_X) =: g(X)$  : irregularity

⑤  $H^2(X, G_m) : G_m$ -gerbes ( $\cong \text{PGL}_n$ -torsor)

$\rightsquigarrow p_g = \dim H^2(X, \mathcal{O}_X)$

$= \dim H^0(X, K_X)$  geom. genus.

$G_m = \text{Spec } \mathbb{Z}[t, t^{-1}]$  : inv, elem

$\text{Pic}(X)$

$NS$

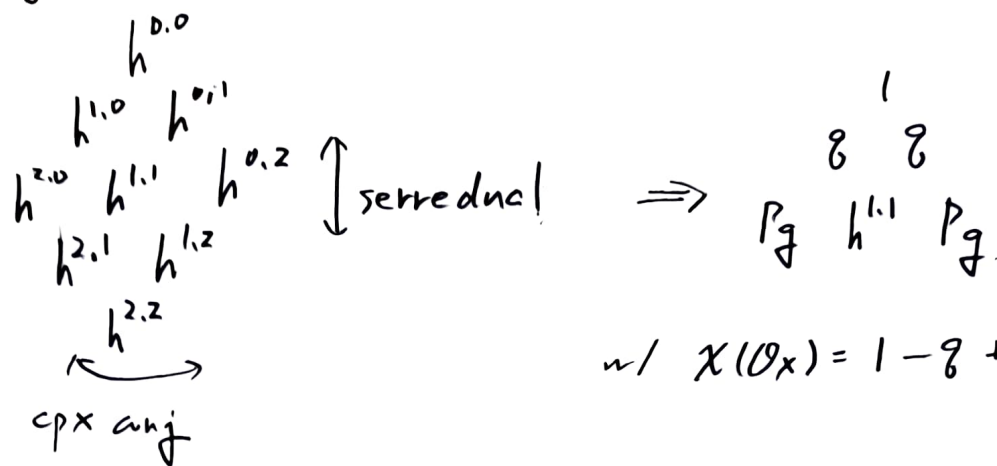
$\cdot \text{alg!}$   
 $\cdot \text{discrete}$

$\text{Pic}^0$

$\cdot \text{conti!}$   
 $\cdot \text{transcend.}$

© Hodge diamond.

$$h^{p,q} = \dim H^q(X, \Omega_X^p)$$



rational

$$\begin{array}{ccc} & 1 & \\ 0 & & 0 \\ & h^{1,1} & 0 \end{array}$$

$$\mathbb{P}^1 \times \mathbb{P}^1 \Rightarrow h^{1,1} = 1 + 1$$

$$\mathbb{P}^1 \times \mathbb{P}^1 \Rightarrow h^{1,1} = 2$$

Note!

$h^{1,1}$ : NOT bir inv.  
(def inv)

ruled

$$\begin{array}{ccc} g & 1 & \\ g & & g \\ 0 & h^{1,1} & 0 \end{array}$$

$$\mathbb{P}^1 \times \mathbb{P}^1 \Rightarrow h^{1,1} = 2 + r$$

K3

$$\begin{array}{ccc} & 1 & \\ 0 & & 0 \\ 1 & 20 & 1 \end{array}$$

Enr

$$\begin{array}{ccc} & 1 & \\ 0 & & 0 \\ 0 & 10 & 0 \end{array}$$

Ab

$$\begin{array}{ccc} & 1 & \\ 2 & & 2 \\ 1 & 4 & 1 \end{array}$$

biellip

$$\begin{array}{ccc} & 1 & \\ 1 & & 1 \\ 0 & 2 & 0 \end{array}$$

⚠  $P_g$  and  $g$  alone do NOT characterize a surface  
one needs information on the canonical class  $K_X$   
(or plurigenera)

Def  $P_m = h^0(X, mK_X)$

$$P_1 = P_g$$

Recall

$$K(X) = \begin{cases} -\infty & P_m = 0 \quad m \gg 0 \\ 0 & P_m: \text{const or periodic} \\ 1 & P_m \sim cm^1 \\ 2 & P_m \sim cm^2 \end{cases}$$

• Castelnuovo  $P_2 = 0, g = 0 \Leftrightarrow$  rational

• Enriques  $P_{12} = 0 \Leftrightarrow$  ruled

• Enriques  $P_1 = P_2 = 1, g = 2 \Leftrightarrow$  Abel surf

see Also: Chen-Hacon "characterization of abelian varieties"

### §3 K3 surf

$K_X$ : triv &  $b_1 = 0$ .

$\leadsto H^0(X, \Omega^2) = \langle \omega_X \rangle$   $\omega_X$ : non-vanishing hol 2-form

④ Fact (See [BHPV] VIII 8)

| all K3-surf are deformations of each other |

• Bogomolov - Tian - Todorov & Torelli

$\Rightarrow$  diffeomorphic.

•  $\sim$  def quartic  $\subset \mathbb{P}^3 \Rightarrow$  all K3 are simply conn.

⑤ Noether formula:  $\chi(\mathcal{O}_X) = \frac{1}{12}(C_1^2 + C_2)$

$\Rightarrow 1 - 0 + 1 = \frac{1}{12}(0 + C_2) \Rightarrow e(X) = C_2 = 24$

⑥ Hirzebruch signature Thm:  $b_+ - b_- = \frac{1}{3}(C_1^2 - 2C_2) = -16$

$\Rightarrow (H^2(X, \mathbb{Z}), \langle, \rangle)$ : unimodular  
 $\uparrow$  cup prod. (3.19)

• even (hard to show)

$\leadsto H^2(X, \mathbb{Z}) = U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$   $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, E_8(-1) = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$

⑦ observation

$\langle \omega_X, \omega_X \rangle = 0, \langle \omega_X, \bar{\omega}_X \rangle > 0$

$\leadsto \text{Re}(\omega_X)$  and  $\text{Im}(\omega_X)$  generates pos. def 2-dim subsp  $E(\omega_X)$  in  $H^2(X, \mathbb{R})$

$H^{1,1}(X, \mathbb{R}) := E(\omega_X)^\perp$  : (1.19)

Note  $\text{Pic}^0(K3) = \{0\}$

we need to consider  $\cap H^2(X, \mathbb{Z})$

$\Rightarrow NS(X) = H^2(X, \mathbb{Z}) \cap H^{1,1}(X)$  Neron-Severi group  
(Picard lattice)

⑧ Picard number

$0 \leq \rho = \text{rank } NS \leq 20$



## § Interlude @ Polarized K3

$L$ : ample w/  $L^2 = 2g - 2$  genus of K3.

( $L$ : ample  $\Rightarrow |3L|$ : very ample) See Saint-Ponsat & Hubrehtz.

$g = 2$ :  $X \xrightarrow{2:1} \mathbb{P}^2$  ramified on a sm sextic.

$$X \subseteq \mathbb{P}(1,1,1,3)$$

$g = 3$ :  $(4) \subset \mathbb{P}^3$

$g = 4$ :  $(2,3) \subset \mathbb{P}^4$  complete intersection.

$g = 5$ :  $(2,2,2) \subset \mathbb{P}^5$

$g = 6 \sim 10$ : Mukai model Grass  $\cap \mathbb{P}^9$

## @ Kummer surf.

$$A = \mathbb{C}^2 / \Lambda : \text{cp}^x \text{ tori } \hookrightarrow (-1)A$$

16 torsion pts as fixed pt  $\frac{1}{2}\Lambda/\Lambda \leadsto A_1$ -sing

$\leadsto$  min resol gives  $\text{Kum}(A)$

elliptic K3. @ lattice polarized K3 @ Singular K3.  
(Mirror symm) (Shioda-Inose)

## §4 Lattice theory

$P(X) = \{x \in H^{1,1}(X, \mathbb{R}) : \langle x, x \rangle > 0\}$

$P^+(X) \ni$  Kähler class. : positive cone

$\Delta^+(X) = \{\delta \in N_S : \text{eff } \langle \delta, \delta \rangle = -2\}$  : root

$H_\delta = \{x \in P^+(X) : \langle x, \delta \rangle = 0\}$  : wall

$W(X) = \{s_\delta(x) = x + \langle x, \delta \rangle \delta\}$  : reflection gp

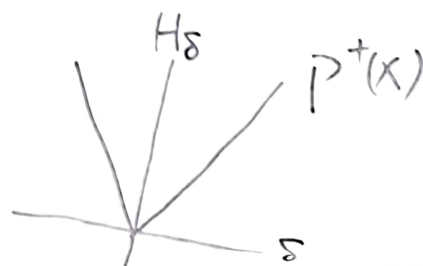
$D(X) = \{x \in P^+(X) : \langle x, \delta \rangle > 0 \ \forall \delta \in \Delta^+(X)\}$  fund. dom  $\hookrightarrow W(X)$

$\Rightarrow D(X)$  : Kähler cone (or ample cone  $(\cap \mathbb{Z})$ )



$$x^2 = x_1^2 - x_2^2 - \dots - x_n^2$$

$$x_i \geq 0 \quad (2 \text{ up})$$



Thm ([BHPV] Thm 11.1, [Kon] Thm 6.1)

[6]

Suppose  $\phi: H^2(X, \mathbb{Z}) \rightarrow H^2(X', \mathbb{Z})$  isom.

s.t.  $\bullet \phi(W_X) \in CW_{X'}$

$\bullet \phi(D(X)) = D(X')$

$\Rightarrow \exists! \psi: X' \rightarrow X$  isom w/  $\psi^* = \phi$

idea of pf ① local Torelli: period map is loc. isom. (deformation theory)

② Torelli for Kummer

using Torelli for complex tori (affine geometry)

③  $\{\text{period of Kum}\} \subset_{\text{dense}} \{\text{period of K3}\}$

④ period of Kum  $\rightsquigarrow$  converge period of K3

① period map.  $L = U^{\oplus 2} \oplus F_8(-1)^{\oplus 2}$

$\Omega := \{w \in P(L \otimes \mathbb{C}) : \langle w, w \rangle = 0, \langle w, \bar{w} \rangle > 0\}$  : period domain

(IV<sub>n</sub>-type bounded symm dom)

i.e.  $\text{Im}(z_1)^2 - \sum_{i=2}^n \text{Im}(z_i)^2 > 0$ .

$\alpha_X: H^2(X, \mathbb{Z}) \xrightarrow{\sim} L$  marking

$(X, \alpha_X)$  : marked K3.

$\alpha_X(W_X)$  : period.

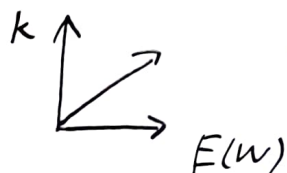
$\mathcal{M} := \{\text{marked K3}\} / \sim$

$\lambda: \mathcal{M} \rightarrow \Omega$  period map.

$(X, \alpha_X) \mapsto \alpha_X(W_X)$

② surjectivity of period map.

local-Torelli + wall crossing



$\langle k, \delta \rangle = 0 \quad \delta^2 = -2 \Rightarrow \text{wall crossing } \delta$

① Moduli

[7]

$\Omega/O^+(L)$  : non-Hausdorff of dim 20.

↗ choice of marking

$\mathcal{F}_{2d} = \Omega_{L^\perp}/T_{2d}$  : quasi-pring of dim 19 (Noether-Lefschetz locus).  
w/o wall

→ Satake-Baily-Borel compactification / quasi-pol.  
(big & nef)

Note  $\mathcal{F}_j \cap \mathcal{F}_{j'} \neq \emptyset$

One K3 may admit various polarization.

## §5 References

### General Theory

- BHPV
- Beauville

### K3 • Huybrechts

- Kondō
- Saintu-Donat

### Enr • Kondō

- Cossec-Polgachev-Liedtke
- Polgachev-Kondō
- Katsura (in char 2)

### Abel • Mumford

- Birkenhake-Lange
- Polishchuck

### Biellip • BHPV

- Beauville